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| **Performance Goal Text:** | To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity | | |
| | The goal of this project is to use recently-developed, high-performance computational methods to study long-standing problems in the theory of superfluidity, and to provide a theoretical foundation for future experiments with ultra cold atomic gases, He-4 superfluid, solid, and supersolid phases. 1. Critical exponents for the XY-universality class. The objective is to perform large-scale simulations of various models with the complex number order-parameter field at the critical point in order to determine the correlation length index with the world-record accuracy (from finite-size and reduced temperature scaling of the superfluid density).<br>2. Critical points and the universal thermodynamics in the fluctuation region of weakly-interacting Bosonic and field-theoretical models in two and three dimensions. The interaction-induced shift of the normal-superfluid transition point and the thermodynamic relations for the condensate, superfluid and particle densities in the fluctuation region are | | |

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| **Task Description:** | universal for all weakly-interacting Bosonic systems. The objective is to construct the corresponding universal scaling functions numerically with high accuracy, and to establish conditions for the experimental studies of the fluctuation region in ultra-cold atomic systems.<br><br>3. Recent experiments by Kim and Chan on the non-classical moment of inertia of solid He-4 have excited a new wave of interest in the suprsolid phase of matter. The situation with supersolidity of He-4 remains controversial, both at the experimental and theoretical levels. The objective is to develop new algorithms for first-principles simulations of large (several thousand atoms) He-4 and other bosonic systems and to study their superfluid properties under various conditions: ideal crystals, grain boundaries, dilocations, polycrystals, generic quenched disorder, etc.<br><br>1. The most accurate up to date MC simulations of Hasenbusch et al. and Campostrini et al. for the critical exponents of the normal-superfluid transition in He-4 are in disagreement with the space flight experiments. This disagreement calls for an independent high-precision determination of critical exponents, especially of the specific heat index which is directly related to the correlation length and superfluid density index. 2. The problem of the interaction-induced shift of the critical temperature in a weakly interacting Bose gas in 3D has attracted much theoretical effort in recent years. Apart from its fundamental interest, the study of this problem has been strongly stimulated by the experimental achievement of Bose-Einstein condensation in ultra-cold gases. Contrary to the 3D case, the field theoretical treatment of the 2D system is notoriously difficult since the physics behind the 2D transition is connected with the renormalized coupling between vortices and the structure of the vortex core, not to mention that the critical region is very wide. Much of the confusion originates from the change of the universality class of critical fluctuations when interactions (no matter small) are switched on.<br><br>3. At present, the best numerical schemes for first-principles simulations of bosonic systems, based on the path-integral representaion of quantum statistics, are very inefficient in calculating superfluid properties in large systems (the record is about 5 particles per dimension). This limits our ability to study the thermodynamic limit, to perform simulations of polycrystalline samples and crystal defects. If simulations can be extended to several thousand atoms, we would be able to answer conclusively such long standing problems in helium physics as the superfluidity of ideal crystals, grain boundaries, dislocations, or polycrystalline matter. In addition, much higher accuracy can be achieved in simulations of the liquid phase, in the bulk and in restricted geometries. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | We have developed a new approach to path-integral Monte Carlo simulations of Bosonic systems such as Helium-4 and hydrogen. It involves two novel features which distinguish it from existing methods (i) our algorithm works with the configuration space of the single-particle Grenn function and partition function at the same time, and allows efficient simulations of off-digonal corelations and grand canonical ensemble properties in systems two orders of magnitude larger then possible previously, (ii) the attractive part of the inter-particle potential is accounted exactly at no computational cost by using diagrammatic expansion into Feynman diagrams. |
| | 1. The hyperscaling hypothesis, which establishes the relation between the specific heat and the correlation length exponents, is the cornerstone of the modern theory of critical phenomena. For several years there was a strong discrepancy between the experimental value of the specific exponent and the high-precision numerical result for the correlation length exponent.<br>In Monte Carlo simulations by Campostrini et al. the correlation length exponent equal to 0.67155(27) was obtained using finite size scaling technique at a special critical point of the model lattice system where leading corrections to scaling were determined to be minimal (point A). Correspondingly, in the final analysis of the data, corrections to scaling were excluded from the fit.<br><br>Our simulations of the critical point A using Worm Algorithm approach indeed show a nearly perfect scaling law for the superfluid density derivative for system sizes L>8. By excluding corrections to scaling from the fit we obtained the corelation length exponent as 0.67185(4), which was in even stronger disagreement with the space-flight experiments (assuming that the hyperscaling relation is valid) due to much smaller error on the answer. However, when simulations were performed at the nearby critical point B we again found a nearly perfect scaling for L>8, but now with the exponent 0.67112(5). It became obvious, that the precision of our simulations was high enough to see leading corrections to scaling at the special point A, neglected by Campostrini et al. With corrections to scaling included into the joint fit of the two sets of data for points A and B we finally determined the correlation length exponent as 0.6717(3), As expected, the error has dramatically increased, but the result was one of the most accurate ever. After our work was published in the condensed matter archive, a new independent study by Campostrini et al. has confirmed all four digits in our result, 0.6717(1).<br><br>In the meantime, the experimental value of the specific heat exponent was reconsidered for the second time (by four standard deviations) and the latest result is -0.0127(3), or, for the correlation length exponent 0.6709(1). It seems that the hyperscaling relation is satisfied with high accuracy and there is no contradiction between the space-flight data and numerical results, but the experimental error bar is severely underestimated.<br><br>2. Universal properties of the 2D Bose gas turned out to be a relatively easy problem numerically. We studied the crossover between the mean-field and critical behavior of the two-dimensional Bose gas throughout the fluctuation region of the Berezinskii--Kosterlitz--Thouless phase transition point, and established universal (for all weakly interacting complex-field models) relations between the thermodynamic parameters of the system, including superfluid and quasi-condensate densities. We have also developed the mean-field theory based on the notion of quasicondensate. It was shown to work remarkably well outside of the fluctuation region which we now can characterize quantitatively. Our results provide the best reference data to compare with and to study interaction effects in superfluid 2D Bose systems.<br><br>An analogous study of the 3D case was far more difficult, as expected. We find that in three dimensions the mean-field theory also has to be based on the notion of quasicondensate, not condensate. The numerical evidence in favor of the quasicondensate theory is overwhelming, and comes as a totally unexpected result in three dimensions. Recently, several theoretical groups attempted non-perturbative calculations of the fluctuation region and critical points where such properties as the critical density, temperature, and critical chemical potential shifts, the superfluid and condensate densities as well as the order parameter correlation function decay are universal. A reliable theoretical scheme should |

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| **Task Progress:** | be able to reproduce all of them, and we provide the corresponding benchmarks in our study. So far, none of the analytic approaches was capable of achieving this goal. We have also computed density profiles for ultra-cold atomic gases in loose traps and found that a significant fraction of the density profile might correspond to the fluctuation region even when the condensate density in the middle of the trap is comparable to the critical density. Surprisingly, interaction effects are strong and are easily detectable for gases with extremely small gas parameter na3, where a is the scattering length.<br><br>3. We have proved that the necessary condition for a solid to be also a superfluid is to have zero-point vacancies, or interstitial atoms, or both, as an integral part of the ground state. As a consequence, superfluidity is not possible in commensurate solids which break continuous translation symmetry. One consequence is that recent experiments by Kim and Chan can not be interpreted as bulk supersolidity. One scenario for the non-classical moment of inertia of solid He-4 is the superfluidity of micro-crystallite interfaces. On the basis of the most simple model of a quantum crystal--the checkerboard lattice solid--we show that the superfluidity of interfaces between solid domains can exist in a wide range of parameters. At strong enough inter-particle interaction, a superfluid interface becomes an insulator via a quantum phase transition. Under the conditions of particle-hole symmetry, the transition is of the standard U(1) universality class in 3D, while in 2D the onset of superfluidity is accompanied by the interface roughening, driven by fractionally charged topological excitations.<br><br>We have developed a new approach to path integral Monte Carlo (PIMC) simulations based on the worm algorithm, originally developed for lattice models and extended now to continuous-space many-body systems. The scheme allows for efficient computation of thermodynamic properties, including winding numbers and off-diagonal correlations, for systems of much greater size than that accessible to conventional PIMC. As an illustrative application of the method, we simulated the superfluid transition of Helium-four in two and three dimensions.<br><br>In a separate work, we study different solid phases of He-4. Our study includes simulations that start off from a high-T disordered phase (at the T=0 melting density of the hcp crystal as well as at high pressure) which is then "quenched". The low-T properties of the system crucially depend on the initial state. While an ideal hcp crystal is a clear-cut insulator, the disordered system quenched down to T=0.2 K freezes into a superglass, namely a metastable amorphous solid phase, that features off-diagonal long-range order and superfluidity.<br><br>Several theoretical proposals link supersolid properties in Helium-4 to vacancies. We investigated various scenarios of how equlibrium or out-of-equilibrium vacancies can result in superfludity of Helium-4 crystals. Accurate results for the vacancy and interstitial formation energies were reported, with the conclusion that these defects are thermally activated at all pressures and thus can not play any role in the ground state. We also demonstrated that a dilute homogeneous gas of vacancies in a hexagonal-close-packed crystal is thermodynamically unstable against phase separation at low temperature, due to anisotropic elastic interactions among vacancies. A cluster of vacancies in an otherwise perfect crystal is formed and already three vacancies were a very tight bound state. These simulations rule out all vacancy-related mechanisms of supersolidity in Helium-4.<br><br>In a related work, we establish the nature of the supersolid phase observed for hardcore bosons on the triangular lattice near half-integer filling factor, and study the phase diagram of the system at finite temperature. We find that the solid order is always of the (2m,-m',-m') with m changing discontinuously from positive to negative values at half-filling, contrary to predictions of other phases, based on an analogy with the properties of Ising spins in transverse magnetic field. At finite temperature we find two intersecting second-order transition lines, one in the 3-state Potts universality class and the other of the Kosterlitz-Thouless type. |

| **Bibliography Type:** | Description: (Last Updated: 12/27/2006) |
|---|---|
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